Andhra University College of Engineering

Electronics and Communication Engineering | Course Code: EC301

Digital Signal Processing - Laboratory Manual & Practical Guide

Syllabus Focus: EC301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Discrete Fourier Transform & FFT

" Syllabus Scope: Properties of DFT, Circular convolution, Radix-2 Decimation-In-Time DIT and Decimation-

Unit 2: Infinite Impulse Response IIR Filters

" Syllabus Scope: Analog filter approximations Butterworth, Chebyshev, Bilinear transformation method, Im

EC301: Digital Signal Processing - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Finite Impulse Response FIR Filters

" Linear phase FIR filter characteristics, Symmetric and antisymmetric impulse response, Windowing techniques

Unit 4: Realization of Digital Filters

" Direct Form I, Direct Form II, Cascade, Parallel, and Lattice ladder realizations for IIR and FIR filter arc

Unit 5: Finite Word Length Effects

" Fixed-point and floating-point representations, ADC quantization noise, Coefficient quantization, Round-off

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Computation of N-Point DFT and IDFT and Verification of Circular Convolution in MATLAB/Python

Lab 2: Implementation of Radix-2 Decimation-in-Time FFT Algorithm

Lab 3: Design and Magnitude Response Verification of IIR Butterworth Low-Pass and High-Pass Filters

Lab 4: Design of Linear Phase FIR Filters using Rectangular and Hamming Windows

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Alan V. Oppenheim, Ronald W. Schafer - Discrete-Time Signal Processing Pearson 3rd Ed

* John G. Proakis, Dimitris G. Manolakis - Digital Signal Processing Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students